

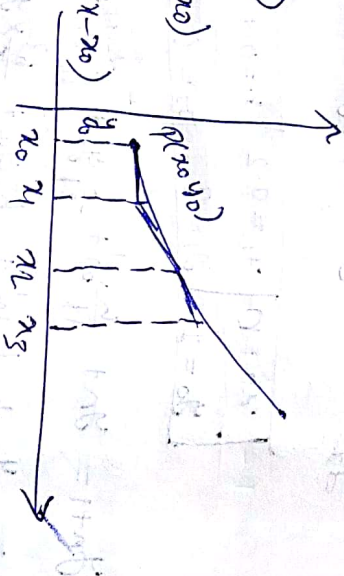
Euler's method

Standard form:-

Solve, $\frac{dy}{dx} = f(x, y)$ given $y(x_0) = y_0$.

$$y - y_0 = \text{slope} (x - x_0) \\ = f(x_0, y_0) (x - x_0)$$

$$y = y_0 + f(x_0, y_0) (x - x_0)$$



$$y_1 = y_0 + f(x_0, y_0) (x_1 - x_0)$$

$$y_1 = y_0 + h f(x_0, y_0)$$

Similarly

$$y_2 = y_1 + h f(x_1, y_1)$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Here to solve $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$

let us take the points $x_0, x_1, x_2, \dots$ in equal intervals

let the actual solⁿ of the differential eqⁿ denoted by the graph (continuous line graph) ~~the~~ $P(x_0, y_0)$ lies on the curve.

then we have to cal. $y_1, y_2, \dots$

now eqⁿ of tangent at the point x_0, y_0 is

$$y - y_0 = \text{slope} (x - x_0)$$

$$y - y_0 = f(x_0, y_0) (x - x_0)$$

$$y = y_0 + f(x_0, y_0) (x - x_0)$$

hence, we can calculate

∴ solve the D.E. $y' = x - y^2$ & $y(0) = 1$ by using Euler's method. Hence cal. $y(1)$ & $y(2)$.

$$\frac{dy}{dx} = x - y^2$$

$$h = 0.1$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Improved Euler's Method

General formula

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_n + hf(x_n, y_n))]$$

Put $n=0$, we get.

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0))]$$

Similarly $\rightarrow$

Ex 1 generally $y_2 = \dots$

Ex 2 solve the eqⁿ $5x \frac{dy}{dx} + y^2 = -2 = 0$; $y(4) = 1$

find $y(4.1)$ & $y(4.2)$ taking $h = 0.1$ using

improved Euler's method.

Case (i) :-

Range-Kutta (R-K) method (4th order).

Ex 1 given :-

$$\text{To solve } \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

find

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2})$$

$$k_3 = hf(x_0 + h, y_0 + k_2)$$

$$k_4 = h f(x_0 + h, y_0 + k_3)$$

$$\Delta y = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\text{where } \Delta x = h$$

$$y_1 = y_0 + \Delta y$$

Q. Obtain the values of y at $x = 0.1, 0.2$ using 4th order RK method for the D.E

$$y' = -y, \quad y(0) = 1$$

$$\text{given, } y' = f(x, y) = -y$$

$$y(0) = 1 \Rightarrow x_0 = 0$$

$$y_0 = 1$$

$x_0 = 0$	0.1	0.2
$y_0 = 1$	$y_1 = ?$	$y_2 = ?$

$$y_1 = y_0 + \Delta y$$

$$\Delta y = \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = h f(x_0, y_0)$$

$$f(x, y) = -y$$

$$= 0.1 \times (-1)$$

$$f(x_0, y_0) = -y_0 = -1$$

$$= -0.1$$

$$k_2 \Rightarrow h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$\Rightarrow 0.1 f\left(0 + \frac{0.1}{2}, 1 + \frac{(-0.1)}{2}\right)$$

$$\Rightarrow (0.1) f(0.05, 0.95)$$

$$\Rightarrow -0.095$$

$$k_3 = h f(x_0 + h, y_0 + k_2)$$

$$= 0.1 f(0.1, 0.9525)$$

$$= -0.09525$$

$$k_4 = h f(x_0 + h, y_0 + k_3)$$

$$= (0.1) f(0.1, 0.90475)$$

$$\Rightarrow -0.090475$$

$$k_1 = -0.08186517$$

$$\Delta y = -0.086507 - (-0.08186517)$$

$$y_2 = y_1 + \Delta y = 0.81873089$$

Assignment Qn

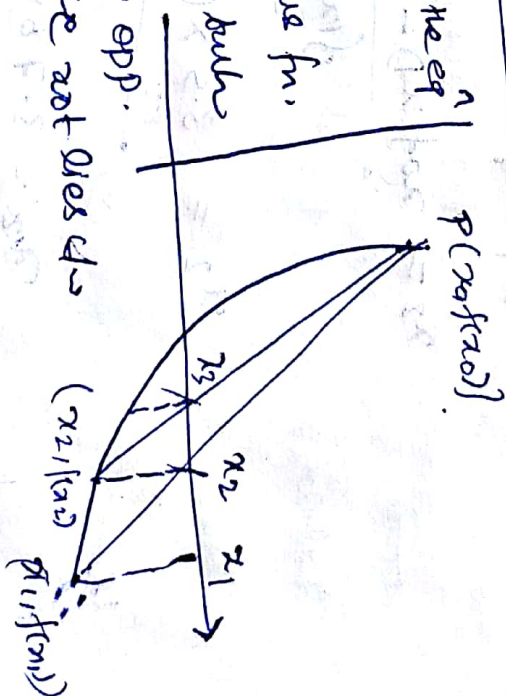
Qn 2nd order RK algorithm with one example.

Method of False position or Regula-falsi

Aim: - To find the root of the eq

$$f(x) = 0$$

Let $y = f(x)$ be a continuous fn, in the interval (x_0, x_1) such that $f(x_0) \& f(x_1)$ are in opp. signs. It implies that the root lies b/w $x_0 \& x_1$.



Now eqⁿ of the line PQ is $\rightarrow$

$$\frac{y - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$y - f(x_0) = \frac{[f(x_1) - f(x_0)]}{(x_1 - x_0)} (x - x_0) \rightarrow \textcircled{1}$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

The procedure is repeated till the root is found to the required accuracy.

∴ find the root of eqⁿ $2x - \log_{10} x = 7$. which lies b/w 3.5 & 4. correct to 5 decimal using regula falsi in 2 iterations. Answer

Solⁿ $x_0 = 3.5$
 $x_1 = 4$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{3.5 \times (2 \times 4 - \log_{10} 4 - 7) - 4 \times (2 \times 3.5 - \log_{10} 3.5 - 7)}{0.39794}$$

$$x_2 = 3.78878$$

$$f(x_2) = 2 \times 3.78878 - \log_{10} 3.78878 - 7$$

$$= -9.39388 \times 10^{-1}$$

∴ Root lies b/w 3.78878 & 4

Take $x_0 = 3.78878$
 $x_1 = 4$

$$x_3 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{(3.78878)(0.39794) - 4(-9.39388 \times 10^{-1})}{0.39794 - -9.39388 \times 10^{-1}}$$

$$= 3.78927$$

∴ find the method of Regula falsi a root of keyⁿ $x^3 + x^2 - 3x - 3 = 0$ lying between 1 & 2. upto 3 iterations.

Let $x_0 = 1, x_1 = 2$

$$f(x_0) = 2 - 3 - 3 = -4$$

$$f(x_1) = 8 + 4 - 6 - 3 = 3$$

∴ approximation then root must lie b/w 1 and 2

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{1 \times 3 - 2 \times (-4)}{-4 - 3} = \frac{-4 - 6}{-7} = \frac{-10}{-7} = 1.42857$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Q. find a real root of $x^3 - 3x + 1$ by Newton-Raphson method.

$$f(x) = x^3 - 3x + 1$$

$$f'(x) = 3x^2 - 3$$

3rd iteration

$$f(1) = 1^3 - 3(1) + 1 = -1$$

$\therefore$ a real root is $x = 1$

$\therefore$ Take $x_0 = 0, x_1 = 1$

$$f(x) = x^3 - 3x + 1$$

$$f'(x) = 3x^2 - 3 \Rightarrow f'(x_0) = 3(0)^2 - 3 = -3$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\text{put } n=0, x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f'(x_0)$$

$$x_1 = 0 - \frac{-1}{-3} = -\frac{1}{3}$$

$$x_2 = -\frac{1}{3} - \frac{f(-\frac{1}{3})}{f'(-\frac{1}{3})}$$

Take $x_0 \in [0, 1]$

$$x_1 = \frac{x_0 \sin x_0 + \cos x_0 + 1}{3 + \sin x_0}$$

Take $x_0 \in [0, 1]$

(you can take any value.)

$$x_1 = \frac{(0.6) \sin(0.6) + \cos(0.6) + 1}{3 + \sin(0.6)}$$

$$x_1 = 0.6071$$

$$x_2 = \frac{x_1 \sin x_1 + \cos x_1 + 1}{3 + \sin(0.6071)} \approx 0.6071$$

Since, $x_1 = x_2 = 0.6071$, a root of $f(x) = 0$ is 0.6071 .

Q. find the root of $x = e^{-x}$ by NRM.

(carryout two iterations).

So: Hence, eqⁿ is $x = e^{-x}$

OR

$$x = \frac{1}{e^x}$$

$$\Rightarrow x e^x - 1 = 0$$

$$f(x) = x e^{x-1} - 1$$

$$f'(x) = (1+x)e^{-x}$$

$$f(0) = -1 < 0$$

$$f(1) = e^{-1} > 0$$

$\therefore$ a root is lies b/w 0 & 1.

Hence choose $x_0 = 0.5$

$$\text{By NRM, } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= \frac{x_n^2 e^{x_n} + 1}{(1+x_n) e^{x_n}}$$

$$n=0 \Rightarrow x_1 = \frac{x_0^2 + e^{x_0} + 1}{(1+x_0) e^{x_0}} \quad \text{Start with } x_0 = 0.5$$

$$x_1 = \frac{0.5^2 + e^{0.5} + 1}{(1+0.5) e^{0.5}}$$

$$= 1$$

$$x_2 =$$

Gauss-Seidel Iteration method

Consider the system of eqⁿ:-

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

The eqⁿ can be written as

$$x = \frac{1}{a_1} [d_1 - b_1y - c_1z] \rightarrow \textcircled{1}$$

$$y = \frac{1}{b_2} [d_2 - a_2x - c_2z] \rightarrow \textcircled{2}$$

$$z = \frac{1}{c_3} [d_3 - a_3x - b_3y] \rightarrow \textcircled{3}$$

Then take initial approximations as ~~$x = x_0, y = y_0, z = z_0$~~

$$\text{or } y = y_0, z = z_0 \text{ in } \textcircled{1}$$

$\therefore$ we get first approximation value $\rightarrow$

$$x_1 = \frac{1}{a_1} [d_1 - b_1y_0 - c_1z_0]$$

put $x = x_1$ & $z = z_0$ in $\textcircled{2}$, we get y

$$y_1 = \frac{1}{b_2} [d_2 - a_2x_1 - c_2z_0]$$

$$\text{Now put } x = x_1, y = y_1 \text{ in } \textcircled{3} \Rightarrow z_1 = \frac{1}{c_3} [d_3 - a_3x_1 - b_3y_1]$$

continue the process for better approximation,

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 10z = 25$$

Given eqⁿ are diagonally dominant.

$\therefore$ we write the eqⁿ as

$$x = \frac{1}{20} [17 - y + 2z]$$

$$y = \frac{1}{20} [-18 - 3x + z]$$

$$z = \frac{1}{10} [25 - 2x + 2y]$$

Here choose the initial approximations as

$$y_0 = 0 \text{ and } z = 0$$

$$x_1 = \frac{1}{20} [17] = 0.8500$$

$$\text{put } x = x_1 = 0.85 \text{ \& } z = 0 \text{ in } \textcircled{2}$$

$$y_1 = \frac{1}{20} [-18 - 3x_1 + 50] = \frac{1}{20} [-18 - 3 \times 0.85 + 50]$$

$$= -1.0275$$

then put $x_1 = 0.85$ & $y_1 = -1.0275$ in (3)

we get, $z_1 = \frac{1}{20} [15 - 2x_1 + 3y_1] = 1.0109$.

Second approximation:

Put $y = y_1$ and $z = z_1$ in (1)

$$x_2 = \frac{1}{20} [17 - y_1 + 2z_1] = \frac{1}{20} [17 - (-1.0275) + 2(1.0109)]$$

$$y_2 = \frac{1}{20} [-18 - 3x_2 + z_1] = \frac{1}{20} [-18 - 3(1.0085) + 1.0109]$$

$$= -0.9998$$

$$\text{and } z_2 = \frac{1}{20} [-25 - 2x_2 + 3y_2] = \frac{1}{20} [-25 - 2 \times 1.0085 + 3 \times (-0.9998)]$$

Third iteration:

$$x_3 = \frac{1}{20} [17 - y_2 + 2z_2] = 0.99998$$

$$y_3 = \frac{1}{20} [-18 - 3x_3 + z_3] = -0.995$$